

Lynbrook Math Summer Camp 2026

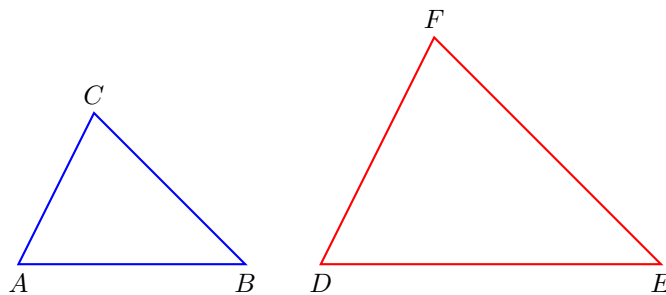
G3 Similarity and Congruence

Lynbrook Math Club

August 2026

G3.1 Lecture

Essentially, we can think of two triangles $\triangle ABC$ and $\triangle DEF$ being similar if the only difference between the two is their size. In other words, one triangle is a “blown up” version of the other, as seen in the diagram below:



In particular, to denote the similarity we write $\triangle ABC \sim \triangle DEF$ (the order of the vertices matters!) and we have the following facts:

- Corresponding angles are equal:

$$\angle ABC = \angle DEF, \angle BCA = \angle EFD, \text{ and } \angle CAB = \angle FDE.$$

- Ratios of corresponding sides is constant:

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}.$$

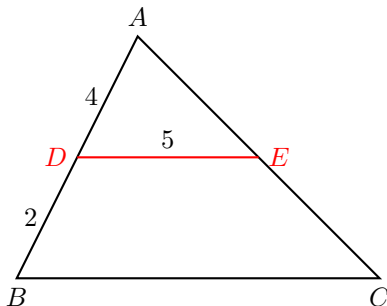
This is pretty powerful, as it will allow us to convert information about angles into information about lengths, and vice versa. But how do we actually know if two triangles are similar?

Angle-Angle (AA) Similarity

If two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar. (Because the angles of a triangle sum to 180° , the third angles must also be equal!)

Example 1

In $\triangle ABC$, line segment DE is drawn parallel to side BC , with D on AB and E on AC . If $AD = 4$, $DB = 2$, and $DE = 5$, what is the length of BC ?



Solution: Because $DE \parallel BC$, we know that $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$ (corresponding angles). Therefore, by AA Similarity, $\triangle ADE \sim \triangle ABC$.

The ratio of their sides must be constant:

$$\frac{AD}{AB} = \frac{DE}{BC} \implies \frac{4}{4+2} = \frac{5}{BC} \implies \frac{4}{6} = \frac{5}{BC}.$$

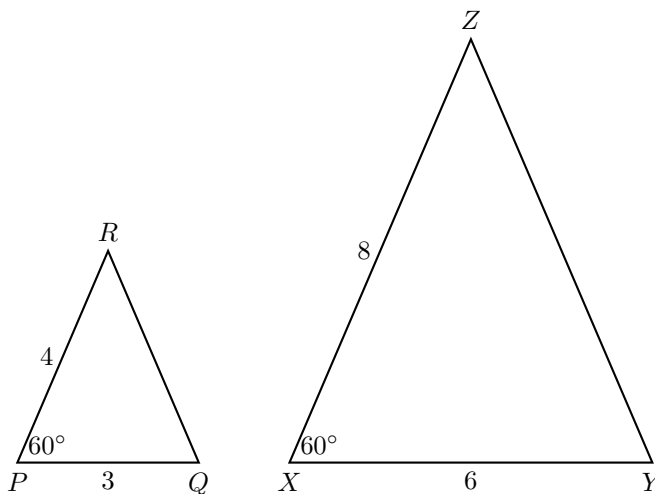
Solving for BC , we get $BC = \frac{30}{4} = 7.5$.

Side-Angle-Side (SAS) Similarity

If two sides of one triangle are proportional to two sides of another triangle and their included angles are congruent, then the triangles are similar.

Example 2

In triangles $\triangle PQR$ and $\triangle XYZ$, you are given that $\angle P = \angle X = 60^\circ$. Furthermore, $PQ = 3$, $PR = 4$, $XY = 6$, and $XZ = 8$. Are the triangles similar? If so, find the ratio of QR to YZ .



Solution: We check the ratio of the sides that include the 60° angle:

$$\frac{PQ}{XY} = \frac{3}{6} = \frac{1}{2} \quad \text{and} \quad \frac{PR}{XZ} = \frac{4}{8} = \frac{1}{2}.$$

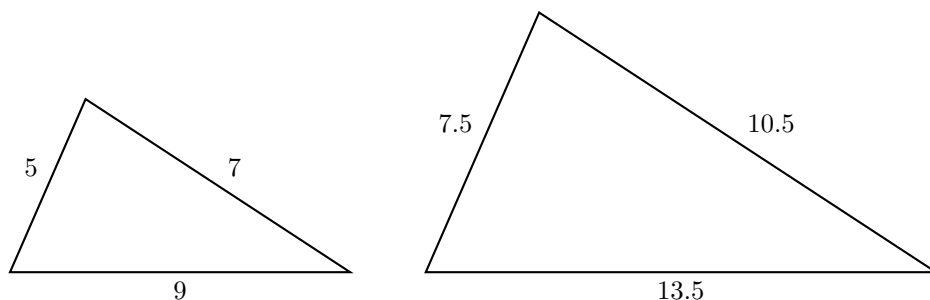
Since the ratios are equal and the included angles are both 60° , $\triangle PQR \sim \triangle XYZ$ by SAS Similarity. The ratio of QR to YZ is therefore also $\frac{1}{2}$.

Side-Side-Side (SSS) Similarity

If all three corresponding sides of two triangles are proportional, then the triangles are similar.

Example 3

A triangle has side lengths 5, 7, and 9. Another triangle has side lengths 10.5, 7.5, and 13.5. Are they similar?



Solution: Let's order the side lengths of both triangles from smallest to largest and check their ratios.

Triangle 1: 5, 7, 9.

Triangle 2: 7.5, 10.5, 13.5.

We check the ratios:

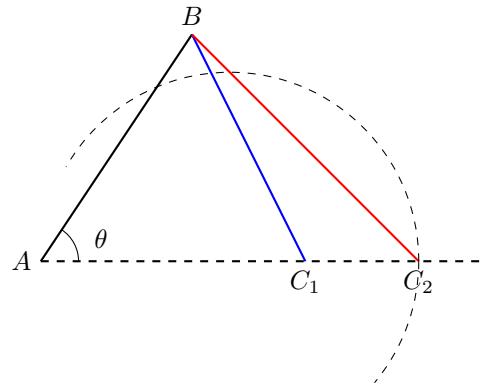
$$\frac{7.5}{5} = 1.5, \quad \frac{10.5}{7} = 1.5, \quad \frac{13.5}{9} = 1.5.$$

Because all three pairs of corresponding sides share the same ratio of 1.5, the triangles are similar by SSS Similarity.

Why SSA Doesn't Work

You might wonder if Side-Side-Angle (SSA) guarantees similarity or congruence. It does **not**. This is often called the *Ambiguous Case*. If you are given the lengths of two sides and a non-included angle, it is possible to draw two completely different, non-similar triangles.

Imagine a triangle where side AB and angle A are fixed, and side BC is given. Think of side BC as a swinging gate on a hinge at B . As it swings, it can hit the line containing AC at two different locations!



In the diagram above, $\triangle ABC_1$ and $\triangle ABC_2$ share the same angle $\angle A$, the same side length AB , and the same side length $BC_1 = BC_2$ (the radius of the circle). However, the two triangles are clearly neither congruent nor similar!